FINTECHATHON 2026 · INTERNATIONAL TRACK · TOPIC F

AI is beginning to control money. Who makes sure it acts correctly?

QED is an AI treasury in which every payment and every trade carries a proof that the blockchain checks before any money moves.

QED

Apurv Chauhan · Lakshya Goswami · St. Xavier's College, Mumbai

AI can decide. QED makes it prove.

THE PROBLEM

AI can follow every rule and still give you a poor rate.

An AI treasury holds dollars and must pay a euro invoice. It converts. Did it get a good price? Today, nothing checks.

01

A minimum is not the best

A slippage floor only says “not worse than X”. A route 33 bps worse than the best still passes a 1% floor.

02

Checked too late

Execution quality is reviewed after settlement, when the money has already moved.

03

Unchecked discretion

The agent decides when and how much to convert, and nobody re-checks the decision.

Brokers owe clients best execution by law. An AI agent owes it to no one.

Routes that clear a 1% floor still give up a median 14–32 bps (1,000 seeded markets, benchmarks/)

THE IDEA · PROOF-CARRYING TRANSACTIONS

Don't trust the agent's decision. Make the chain verify its quality.

1 · PROPOSES

AI agent

Reads the instruction, plans the trade, builds the proof. Holds no key that can pay alone.

→

2 · RE-DERIVES

Independent co-signer

Separate process, own key. Recomputes the decision from the chain; signs only what the owner allowed.

→

3 · VERIFIES

Smart contract

Owner's signed mandate, caps, protected liabilities, and the best-execution proof, in one transaction.

→

4 · OUTCOME

Settles, or reverts

Every check passes and money moves. Any check fails and nothing moves.

> The AI can be wrong, manipulated or stolen. The money still moves only with a valid proof.

THE BREAKTHROUGH · CERTIFIED BEST EXECUTION

The contract checks one inequality, inside the trade itself.

$$\text{output} \geq (1 - \tau) \cdot B$$

B = the most any route through the owner's approved venues could return, from their live reserves. **τ** = one part per million.

1

The owner approves the venues the agent may use.

2

The agent picks any route and sends one price per token as its certificate.

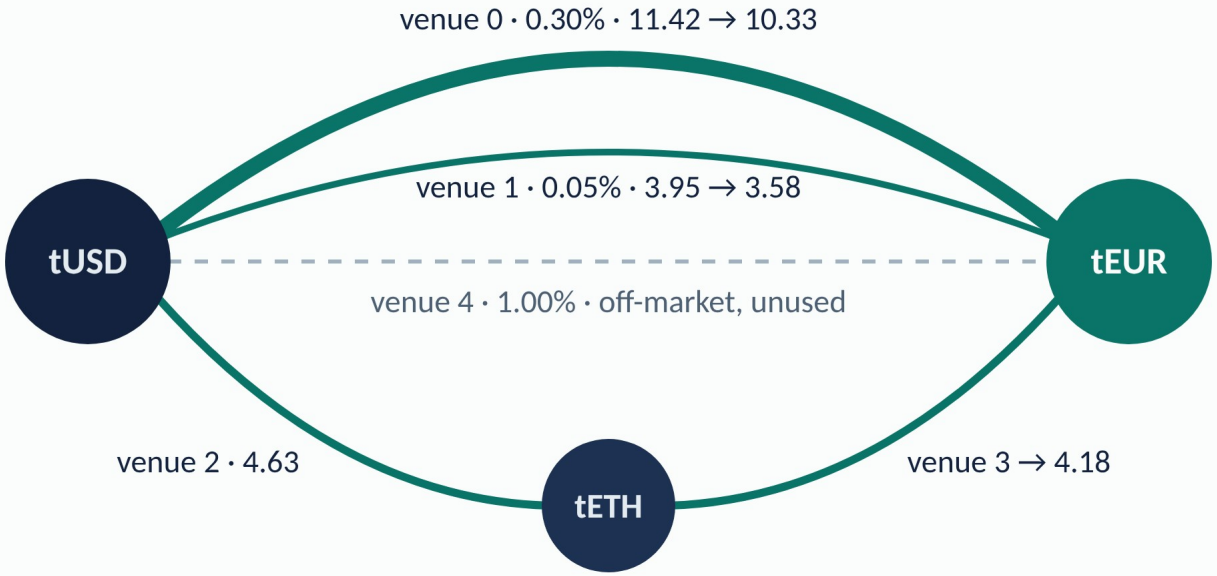
3

The contract recomputes B, rounding against the agent. A worse route reverts.

Why B can't be faked: any prices give a valid upper bound (weak duality). Derivation: appendix A1

THE BREAKTHROUGH, LIVE · SEPOLIA TRANSACTION E1

20 tUSD in, 18.086846 tEUR out. The chain's own bound: 18.086849.



0.17 ppm

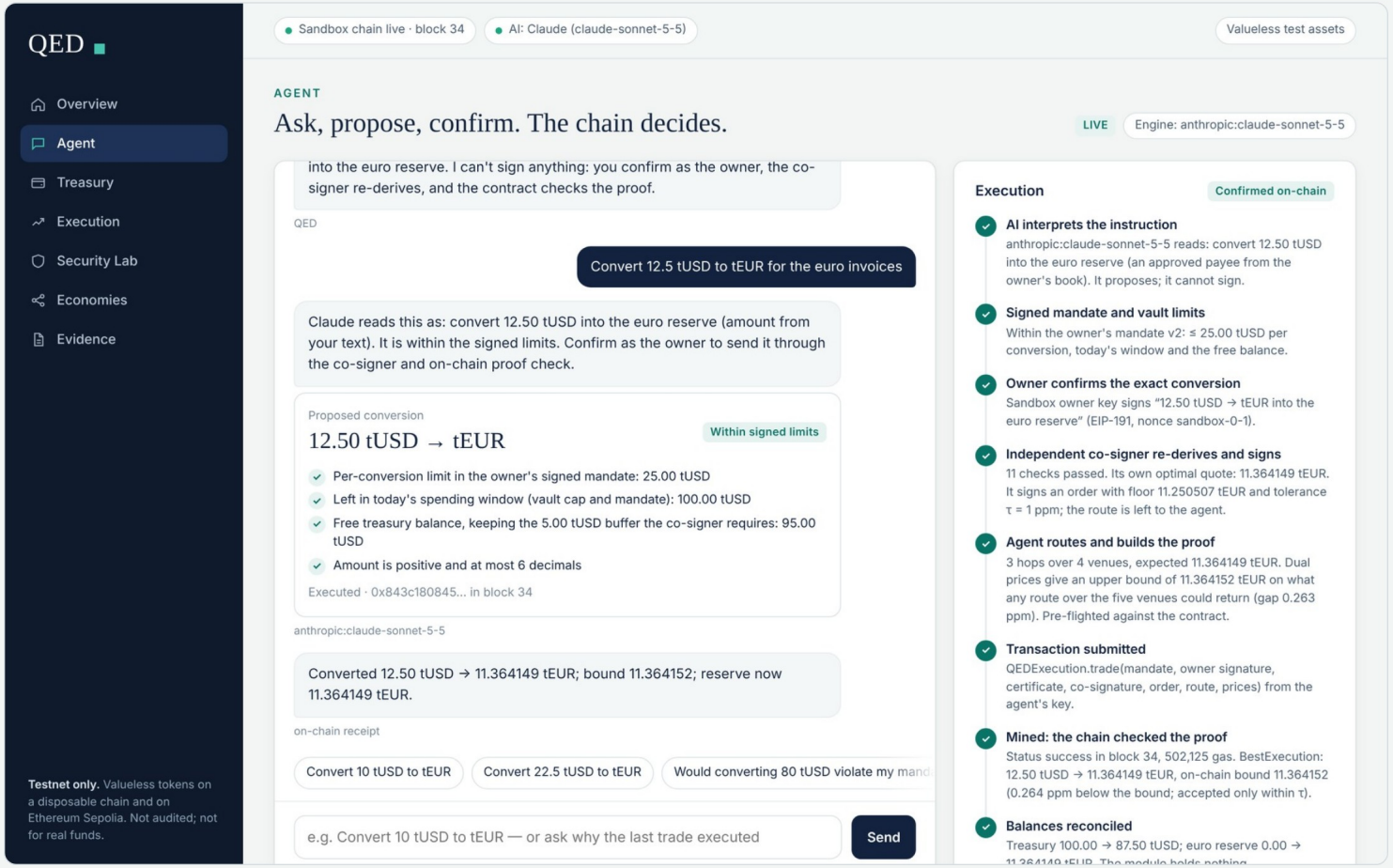
from the most any route through the owner's five venues could have returned

The proof	three prices: v(tUSD) 0.891672 · v(tETH) 2,724.50 · v(tEUR) 1
Checked by	QEDExecution, same transaction
Transaction	`0x56f798cd…9690`

Amounts in tUSD → tEUR · block 11,952,708 · 1,122,781 gas

THE PRODUCT · LIVE NOW

Type an instruction. Watch it get proved.



1 · YOU ASK

"Convert 12.5 tUSD to tEUR for the euro invoices."

2 · THE AI PROPOSES

It shows the signed limits it checked. It cannot sign.

3 · YOU CONFIRM AS OWNER

The co-signer re-derives and signs an order.

4 · THE CHAIN DECIDES

Proof checked, transaction mined, balances reconciled: 0.26 ppm from the bound.

Real QED contracts on a testnet sandbox chain · valueless tokens · no wallet needed · screenshot of the app

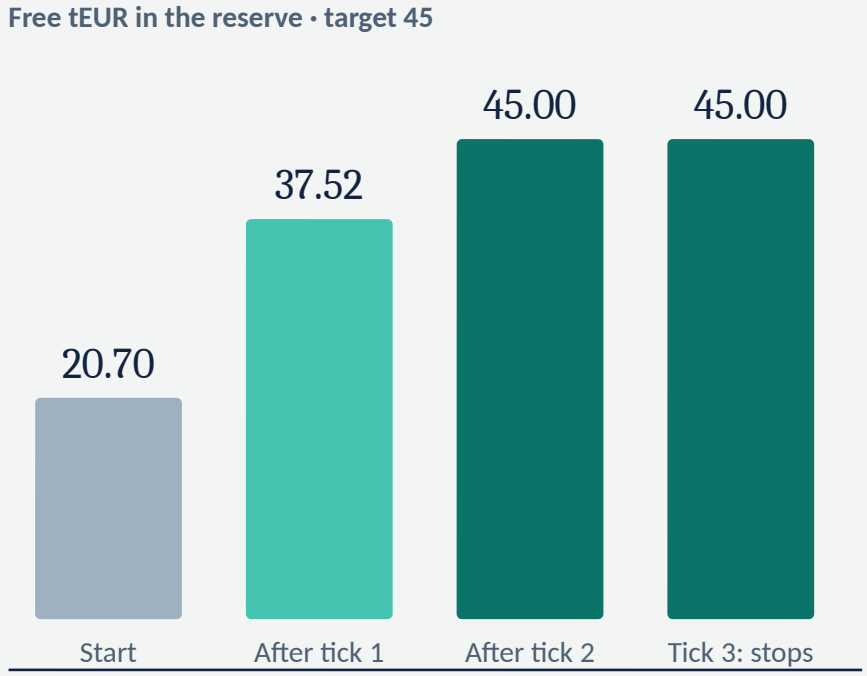
AUTONOMY · SEPOLIA RUNS 4C AND 4D

The owner sets the goal. The agent decides how much, and when to stop.

STANDING POLICY · SIGNED ONCE BY THE OWNER

Keep at least **45 tEUR** free in the euro reserve. Tranches of 2–25 tUSD. At most 150 bps cost each.

Tick 1	Short 24.30 tEUR. Converts 19.54 tUSD, limited by market impact.	PE1
Tick 2	Short 7.48 tEUR. Converts 8.84 tUSD, limited by need.	PE2
Tick 3	Target met. Sends nothing.	



With both hot keys stolen, the on-chain policy still refused over-conversion (QA2) · try it in the app: Treasury

SECURITY UNDER ATTACK · PUBLIC SEPOLIA RECEIPTS

We attacked it with stolen keys. The chain said no.

ATTACK 1

A worse route

A compromised agent routes through the direct pools only: 33 bps below the bound, yet above a 1% slippage floor.

Stopped by QEDExecution · best-execution inequality

`NotBestExecution`

`EA1 · 0xccd07d17…cd9b54937`

ATTACK 2

Pay the attacker

Both hot keys stolen, the agent's and the co-signer's. The attacker tries to pay itself.

Stopped by QEDVault · owner's signed payee list

`PayeeNotInMandate`

`X6 · 0xf9a023d7…29c63c018`

ATTACK 3

Convert too much

Both hot keys stolen again. The attacker converts more of the treasury than the policy needs.

Stopped by QEDTreasuryPolicy · over-conversion clause

`OverConversion`

`QA2 · 0xb49d0c4f…cd23f9f3`

48 attacks on Sepolia, 48 reverts, 0 tUSD to the attacker. Six run live in the app's Security Lab.

Every attack and its custom error: docs/EVIDENCE.md · appendix A12

WHY IT MATTERS · WHO WOULD PAY

Best execution becomes a precondition, not a quarterly report.

Agent treasuries

Businesses, DAOs and AI services that hold one currency and owe another can delegate conversion, including when and how much, without trusting the agent's routing.

Regulated firms

A per-trade record anyone can recompute from public state, for the best-execution duty firms already carry under MiFID II and FINRA rules.

Solvers and intents

A filler can attach the proof and be accepted on it; a protocol can settle on the proof instead of after-the-fact monitoring.

BUSINESS MODEL · A HYPOTHESIS

A verification layer: a fee per certified trade, or a licence for custodians and agent platforms.

NOT YET VALIDATED

No users, revenue or partners yet. Unaudited, testnets only. No market size is claimed.

DEFENSIBILITY · PRIOR ART

Every ingredient is known. The guarantee is not.

Closest work	What it does	What it leaves open
Intents and aggregators: UniswapX, 1inch Fusion, ERC-7683	Enforce a minimum output	Leak up to the tolerance
CoW Protocol, EBBO rule	Compare fills with baseline pools after settlement	Social, after the fact
A^2MM (2021); on-chain one-hop routing (FC 2026)	Compute a split on-chain for parallel pools	Approximate, one pair, no bound
Certified routing (arXiv 2607.22540)	Off-chain KKT certificate for a router	Execution guarded by slippage
Angeris et al., EC 2022	CFMM routing is convex; duality	Theory; routes solved off-chain
QED	**Verifies any route over any owner venues, in the trade's transaction, to 1 ppm**	**Inside an agent's mandate, money on it**

We found no prior system that makes the optimality of an executed route an on-chain condition of the trade.

Sources and the full comparison: docs/PRIOR_ART.md §0 · searched 20 Oct 2026

TEAM AND NEXT STEPS

Built by two students. Ready for a pilot.

Apurv Chauhan

St. Xavier's College, Mumbai

Lakshya Goswami

St. Xavier's College, Mumbai

24
contracts verified on Sourcify

196
real Sepolia transactions

145 + 60
tests passing in CI

WHAT WE ARE LOOKING FOR

1. A treasury team or agent platform to run a pilot with
2. An independent security audit of the contracts
3. Partners to extend the proof to more venue types

Open source: github.com/Nightcore709/qed-agent-wallet

QED

AI can decide. QED makes it prove.

Verifiable financial constraints should be a precondition for autonomous execution, not an audit after the fact.

Try it: the live app, on a testnet sandbox chain (no wallet, no keys)

`qed-judges-preview-production.up.railway.app`

Sepolia proof ledger

`nightcore709.github.io/qed-agent-wallet`

Code, evidence, docs

`github.com/Nightcore709/qed-agent-wallet`

Apurv Chauhan · Lakshya Goswami · St. Xavier's College, Mumbai